

Post Hoc Ergo Propter Hoc?

Side Effect Misattribution and Vaccine Hesitancy

Svenja Miltner*

Jonatan Riberth[†]

Anton Arbman Hansing[‡]

This version: February 2, 2026

Abstract

Side effect concerns are the most frequently reported reason for vaccine hesitancy, yet we lack well-identified evidence on how these risk perceptions form. Using Swedish register data, we study how experiencing illness shortly after vaccination affects the decision to continue vaccinating. We exploit an episode during the COVID-19 vaccination campaign when links between the AstraZeneca vaccine and rare blood clots triggered extensive media coverage and temporary suspensions. Individuals who experience a common blood clot shortly after a dose—clinically unrelated to the vaccine-induced syndrome—drop out of vaccination at almost twice the rate of carefully matched controls (+4pp). Consistent with misattribution, the effect is strongest for diagnoses occurring soon after the dose, and greatly amplified (+16pp) when clinicians report the event as a suspected side effect. Other acute conditions of similar severity produce no comparable response. Strikingly, effects are similarly strong for mRNA vaccine brands, which operate through a different mechanism and were never linked to blood clots. Two groups respond most strongly: younger individuals (who face lower COVID-19 risk) and those who delayed their first dose (indicating baseline skepticism). These findings point to an underappreciated tradeoff: while transparency about vaccine safety is thought to build public trust, such signals may also amplify perceived risks beyond what the evidence warrants.

Keywords: COVID-19, Adverse Events, Vaccine Hesitancy

*FU, Freie Universität Berlin. E-mail: s.miltner@fu-berlin.de.

[†]IIES, Stockholm University. E-mail: Jonatan.Riberth@iies.su.se.

[‡]IIES, Stockholm University. E-mail: Anton.Hansing@iies.su.se.

The authors are grateful to Arash Nekoei, Torsten Persson, Anna-Mia Ekström, Tove Fall, Olof Östergren, Natalia Danzer, Peter Mohr, participants at the 2025 SWECOV workshop and the fall PhD workshop at Freie Universität Berlin. This research was conducted within SWECOV—the Swedish Register-based Research Program on COVID-19—hosted by Stockholm University and financed by Riksbankens Jubileumsfond (RIK21-0004).

1 Introduction

Vaccine hesitancy—the delay or refusal of available and effective vaccines—remains a critical challenge in containing communicable diseases, and is often driven by fears of harmful side effects (Jones, 2020). During the COVID-19 pandemic individuals faced uncertainty: novel vaccines were developed at unprecedented speed and the decision to vaccinate navigated in an environment best described by the World Health Organization as an “overabundance of information, both online and offline”.

Against this background, the transparent communication of vaccine risks becomes a central challenge to pandemic management. Clinicians must report suspected side effects to help detect risks before causality is established, and regulators may impose temporary restrictions or suspensions to limit potential harm while evidence is assessed. These actions are essential for safety monitoring, but they also create salient cues that can be hard for the public to interpret—potentially fueling concerns about vaccine risks and reducing vaccine uptake. As technology allows for increasingly broader and faster spread of information, understanding the impact of the communication of side effects is crucial to carefully carve out efficient pandemic management.

We bring this trade-off to the data by studying how personal exposure to health events after vaccination affects subsequent vaccine uptake. We exploit the context of the AstraZeneca COVID-19 vaccine, that was linked to a rare type of blood clot in the beginning of 2021, accompanied by widespread media coverage and subsequent suspensions of the vaccine in several countries worldwide. Using rich Swedish register data on individual level vaccine decisions and diagnoses, we show that experiencing an unrelated blood clot soon after vaccination substantially increases dropout from the vaccine schedule. Effects are augmented when the blood clot occurs shortly after vaccination and when a clinician reports the event as a suspected side effect. Other acute conditions—whether plausibly vaccine-induced or not—do not show comparable effects.

When people face a novel decision—such as the decision to get vaccinated during the COVID-19 pandemic—they use personal experiences and publicly available information to inform their choice. We hypothesize that the public debate of vaccine side effects shapes how people interpret symptoms after vaccination. In particular, we provide a framework, where highly visible safety signals—such as clinician reporting of suspected side effects or temporary pauses of drugs—shift attention toward vaccine risks and change how people interpret symptoms that occur after vaccination.

Sweden commenced vaccination against COVID-19 in late 2020 and predominantly administered two mRNA vaccines and one adenoviral vaccine by AstraZeneca. In March 2021, reports from several countries raised the suspicion of a possible link between the AstraZeneca COVID-19 vaccine and Thrombosis with Thrombocytopenia Syndrome (TTS), an extremely rare but severe blood clotting condition. Over the course of a few days several countries suspended the AstraZeneca vaccine temporarily for safety concerns. Though the incidence of TTS was no higher than about 10 per million administered doses, media coverage could have contributed to public skepticism about vaccine safety related to blood clots more generally—conditions affecting upwards of 0.5% of the Swedish population annually. AstraZeneca had been used in Sweden for three months by the time of vaccine suspension on March 16th, making up for a total of 200,000 doses administered. The suspension was eventually lifted and vaccination resumed on March 25 for individuals above the age of 65, until the vaccine was finally phased out by July 2021.

Our empirical strategy compares individuals with similar COVID-19 vaccine history, where some experienced a blood clot soon after vaccination while others did not. The credibility of this design stems from the close similarity achieved by pairing each treated individual with a control who received the same vaccine brand(s) in the same month(s), and who is otherwise similar in demographic, socioeconomic, and health characteristics that may shape both blood-clot risk and subsequent vaccine uptake.

Our findings suggest that individuals generalize signals about specific but salient adverse events to other similar diagnoses and drugs. In particular, we find that experiencing a blood clot—unrelated to the COVID-19 vaccine—within six weeks of COVID-19 vaccination reduces uptake of the next dose by 4 percentage points. Given a baseline uptake of 94.5%, this implies a doubling of the dropout risk from the vaccination program. We conclude that the distinct pathology of vaccine-induced blood clot does not prevent individuals from attributing unrelated symptoms to the vaccine they received. We also find moderate spillover effects on close family members. When we combine the response of the diseased individual with these spillovers, roughly half of the overall increase in vaccine hesitancy comes from the individual who experienced the blood clot, while the other half comes from close family members who subsequently discontinue vaccination. We find little support that a general misattribution mechanism could be at play, where unrelated acute events are routinely attributed to recent vaccination. In particular, for other acute conditions that could plausibly have been attributed to the vaccine, we find only very modest effects on vaccine schedule adherence. By contrast, the response to blood clots is sizable, consistent with misattribution that is activated by external cues through intense public attention.

We find that effect size is greatest for blood clots suffered soon after the vaccination. We argue that this is explained by individuals being more likely to draw the causal attribution of the link, rather than incapacitation. Furthermore, when a clinician reports the blood clot event as a side effect of the vaccination—a choice that is anecdotally often communicated to the patient—the vaccine uptake is reduced by a further 13 percentage points. The total effect on these individual is a close to five-fold increase in dropout rates compared to matched controls. Though we can't rule out some selection into reporting, this suggests that clinician communication—much like media coverage—cues individuals to believe that the side effect was real.

Strikingly, we find that the effect is of similar magnitude across vaccine brands, despite no suspected link for vaccines other than AstraZeneca. We infer that, even if prevailing information suggests a specific vaccine-adverse event relationship, individuals do not discriminate across vaccine brands in their risk perception of adverse events when deciding to complete their vaccine schedule.

We proxy perceived vaccination benefits using patients' age and how early they were vaccinated relative to when the vaccine became available, and find smaller effects among individuals with higher expected benefits from vaccinating. This pattern is consistent with vaccination decisions reflecting a trade-off between perceived benefits and perceived costs in the form of side effects. It is less consistent with alternative explanations in which uptake falls mechanically due to incapacitation, or in which the blood-clot episode triggers a non-compensatory reaction such as generalized distrust under which high-benefit individuals would be no less likely to drop out.

Finally, we find no heterogeneity with respect to several proxies for health literacy. Individuals with higher health literacy are equally likely to attribute blood clots to the vaccine they received.

Our paper speaks to both behavioral economics and public health. First, we relate to a literature on how individuals use past experiences in decision making ([Ashraf et al., 2024](#); [Malmendier et al., 2021](#)). We also connect to recent work studying vaccination decisions through this lens ([Bordalo et al., 2022](#)). We contribute by focusing on high-stakes choices during a pandemic and by documenting how salient adverse events, shaped by informational cues, spill over across health domains.

Second, we contribute to a literature on how individuals infer causal relationships from the temporal proximity of events, even when the evidence of an underlying link is weak or absent. While well established in cognitive psychology—often described as an “illusion of causality”—this phenomenon and its consequences have received comparatively little attention in economics. A notable exception is [Malmendier and Tate \(2005\)](#), who show how CEOs misattribute firm performance to their own actions. In a similar vein, [Espín-Sánchez et al. \(2023\)](#) find that churches strategically time prayers to periods when rainfall is more likely, causing the congregation to attribute rainfall to prayer rather than to natural causes. Our focus is on the extent of such misattribution in the healthcare domain. This is closely related to nocebo responses in the medical literature, where expectations about adverse effects lead individuals to experience (psychosomatic) symptoms even in the absence of an active ingredient. Our setting differs in that we study causal attribution following salient realized adverse events, rather than symptoms generated by expectations alone. That said, expectations may still shape attribution: individuals with negative expectations about a vaccine may be more likely to interpret a realized health event as vaccine-related. Consistent with the broader role of attribution and expectations in health, a substantial share of participants in the placebo arms of COVID-19 vaccine trials report adverse events following vaccination ([Haas et al., 2022](#)). Pertinently, [Asan et al. \(2024\)](#) show that patient reports of severe headache (a common presentation of nocebo symptoms) increased substantially after media coverage of rare vaccine-induced blood-clot cases and the associated warning symptoms in Germany.

We are not the first to study how reports linking the AstraZeneca COVID-19 vaccine to blood clots shaped vaccine attitudes during the pandemic. [Agosti et al. \(2022\)](#) and [Deiana et al. \(2022\)](#) analyze vaccine attitudes around the suspension of AstraZeneca. Using aggregate-level data across Europe, they provide evidence that the episode, and particularly the subsequent retraction, fueled skepticism extending beyond the AstraZeneca vaccine itself. While these studies document cross-vaccine spillovers, our individual-level data allows us to exploit personal exposure to blood clot events as a source of variation in salience.

The paper proceeds as follows. Section 2 provides a brief background on vaccine induced blood clots. Section 3 describes the data, and Section 5 explains the empirical strategy. In Section 6 we document our findings, while Section 7 concludes.

2 Vaccine-Induced Blood Clots

Sweden rolled out its first COVID-19 vaccines in December 2020 and, by February 2021, had deployed vaccines from all three major suppliers in Europe: BioNTech/Pfizer, Moderna, and Oxford–AstraZeneca. Soon after market access the EMA received an unexpected number of reports of severe side effects following vaccination with AstraZeneca. Typically, patients would show an unusually low count of blood platelets, internal bleeding and blood clots forming at uncommon sites of the body, such as the brain’s dural venous

sinuses. By March 16 three Northern European countries suspended AstraZeneca's COVID-19 vaccine Vaxzevria alongside Sweden within a week's time.¹ Pending a formal investigation of the events, the EMA announced that the overall benefits of vaccination still outweighed risks of suffering a rare blood clot event. In consequence, the Swedish health care agency resumed vaccination with AstraZeneca for people aged 65 and over on 25 March. The EMA finally confirmed a causal link between the Vaxzevria vaccine and the extremely rare and potentially fatal condition known as vaccine-induced thrombosis with thrombocytopenia (VITT). Thereafter, demand for AstraZeneca's vaccine in Sweden collapsed, and by May 2021 it was phased out as the rollout continued almost entirely with mRNA vaccines.

While the pathological mechanism of VITT is still under investigation, current studies suggest that VITT is caused by an autoimmune response to components specific to adenoviral vaccines. This is in line with the observation, that the adverse events are predominantly observed in patients vaccinated with J&J and AstraZeneca, but not with common mRNA vaccines. Additional research concluded that the AstraZeneca vaccine is not linked to a general increase in the propensity of thrombotic events overall but specifically elevates the risk of rare TTS.

Note, that the unusual clinical picture of VITT, including diagnosis criteria of platelet formation and widespread locations of clotting, which allowed clinicians to quickly rule out VITT in the vast majority of post-vaccination blood clots once diagnostic criteria for VITT were established. On aggregate the hematological condition was diagnosed in 1:150 000 individuals vaccinated with AstraZeneca. To benchmark this against other common types of blood clots: Pulmonary embolism, another severe clotting event, has an incidence rate of 167:100 000 among the Swedish population (Johansson et al., 2014).

Evidence on who was at risk of developing VITT is limited. UK safety monitoring suggested the event was more common among younger adults (about 1 in 50,000 under 50 vs 1 in 100,000 over 50), and early case reports seemed to involve more women, but later work stresses that these patterns may partly have reflected differential rollout and reporting rather than personal risk factors. Overall, studies have not identified individual characteristics that predicts VITT (Klok et al., 2022).

Experiencing a blood clot does not constitute a medical contraindication to COVID-19 vaccination. For example, public health authorities did not advise individuals with a history of common blood clots, such as deep-vein thrombosis following surgery or during pregnancy, to avoid vaccination. However, individuals who developed a blood clot soon after vaccination with AstraZeneca, were recommended to complete their vaccine routine with an mRNA vaccine. Furthermore, the general scientific consensus is that any vaccination against COVID-19 is still highly beneficial even amidst the risk of general thrombotic events. In particular, studies have shown that the risk of suffering a blood clot is significantly higher following infection than following vaccination (Zhao et al., 2024; Katsoularis et al., 2022).

Our empirical strategy builds on reports of vaccine-related blood clots serving as vaccine-safety cues under broad public attention. The mechanism is an interaction: public reporting is common to all, but personal exposure makes clot-related news more salient and more likely to affect behavior. A necessary precondition is that the controversy received broad public attention. Previous studies have documented a

¹The vaccine held market access authorization for the entire European Union and several other countries. Globally, in March 2021, more than as 20 countries halted the administration of the vaccine within a matter of days.

spike in public interest during the time around the AstraZeneca suspension, using Google searches as a proxy for awareness (Deiana et al., 2022; Agosti et al., 2022), suggesting that the adverse events were part of a public debate.² In Sweden, we observe a similar pattern: Google searches related to blood clots spiked in March, when national health authorities announced the suspension of AstraZeneca (see Figure B1). In Figure B2 we provide further evidence on the attention that the controversy received and that this attention was not specific to the AstraZeneca vaccine. In particular, we plot the number of side effects for the three major vaccines against the number of administrated doses. On the day of suspension and shortly after, reported side effects increased substantially for AstraZeneca and also Pfizer, which is commonly perceived as a safe option and has not come under scrutiny in the public debate.³

3 Data

3.1 Administrative Records

We combine data from several Swedish administrative sources covering the entire Swedish population. The data is accessed through the Swedish Register-based Research Program on COVID-19 (SWECOV). The permission to use the data is obtained from Sweden’s Ethical Review Authority (permit numbers 2021-02225, 2022-013550-02, 2022-06118-02 and 2024-02342-02).

Reported Side Effects We draw on individual-level records of suspected adverse drug reactions (ADRs) reported to Sweden’s national spontaneous-reporting system throughout the COVID-19 pandemic, obtained from the Swedish Medical Products Agency between the years of 2005 and 2022. Each entry contains the date of onset and exact notification dates of a reaction, the reporter type (a health-care professional or the individual), the suspected drug link (ATC-code and drug brand name), MedDRA reaction codes, and the EudraVigilance seriousness classification. While spontaneous systems are known to under-report, they are intended to capture any physiological effect judged unrelated to a drug’s therapeutic purpose.⁴

COVID-19 vaccines The Public Health Agency provides us with data on all COVID-19 vaccinations in Sweden between January 2020 and March 2023. These records include the date of administration for each dose and the vaccine brand and manufacturer, allowing us to construct our measure of COVID-19 vaccine hesitancy.

Deaths, diagnoses & drugs We use the Swedish Death Register to identify individuals who died, either due to their blood clot conditions or because of other reasons. Furthermore, we exploit data on healthcare visits from the patient register administrated by the National Board of Health and Welfare. This data entails detailed information on all specialist care visits within the period of 2005–2022, including the date and diagnoses. We

²The authors use key words such as “thrombosis” and “AstraZeneca vaccine.”

³Note, that although not as prominently discussed in public, the mRNA vaccines Pfizer and Moderna have been linked to rare cases of myocarditis and pericarditis.

⁴See García-Abeijón et al. (2023) and Hazell and Shakir (2006) for systematic literature reviews on the nature of under-reporting of adverse events.

use this data to identify cases of blood clot, as well as other conditions, based on the ICD-10-SE classification.

Socioeconomic & demographic characteristics Finally, we access information from Statistics Sweden’s registers on basic socioeconomic and demographic characteristics, such as age, sex, region, income.

In addition, we rely on three sources to create proxies for health literacy. (1) From the military archive, we gather data from military tests that assess cognitive ability. (2) We use data on educational attainment and occupations to pin down individuals who have a university degree or (3) who have a healthcare practitioner in their family.

3.2 Sample and Key Variables

Blood clots We are interested in the effect of experiencing a blood clot post-vaccination. To this end, we first restrict our sample to individuals who have received at least one COVID-19 vaccine. This leaves us with individuals who have a basic willingness to get vaccinated. We further restrict the sample to individuals who were alive by 2023. We use a broad definition of blood clot diagnoses, including conditions that were highly unlikely to have been caused by the vaccine. Instead, the set of diagnoses included are meant to reflect blood clot related conditions that the individual might have inferred to be due to the vaccine. This allows us to draw conclusions about the generalization of the specific severe side effect of VITT to other blood clot diseases. We consider individuals with a relevant blood clot diagnosis within 42 days of their first or second COVID-19 vaccination.⁵ We exclude individuals diagnosed with a blood clot in the five years preceding the COVID-19 pandemic. This allows our treatment to more closely mirror the actual VITT cases, which were unlikely to occur disproportionately among individuals with a prior history of blood clots. This leaves us with a total of 22,957 individuals who developed a blood clot after COVID-9 vaccination.

In Table 1 we display the diagnoses that we consider in our broad definition of blood clots.⁶ The displayed mortality rates refer to the share of individuals that are excluded from the sample because they pass away at some point after developing blood clot and before 2023.⁷ As evident from column (4) in Table 1 the conditions differ remarkably in fatality but as a fair share of individuals have passed away within 1–2 years of developing either of the blood clots can all be classified as severe conditions. In parts of the analysis we consider spillover of diagnoses to family members and also include family members of individuals that passed away. As pointed out in Section 2, the conditions specified in our broad definition of blood clot did not serve as a contraindication for further vaccination. Individuals were unlikely to be advised against taking the COVID-19 vaccine by medical professionals.

Out of the 22,957 individuals who developed blood clot after COVID-19 vaccination, 443 individuals in

⁵Note, that the interval between the first and second dose was in general roughly six weeks. Hence we suspect that cases with vaccination pauses beyond the common six weeks are a group of selected individuals who are potentially already hesitant. While it is possible to include also individuals who developed blood clot after six weeks, it is hence more difficult to interpret the average effect in this group. Furthermore, the actual cases of vaccine induced blood clot typically happened within six weeks, making the blood clot cases under consideration more similar to vaccine induced blood clot.

⁶Throughout the remainder of the paper, we refer to the diagnoses listed in Table 1 collectively as “blood clots” for expositional convenience, although some of these diagnoses are not blood clots in a strict clinical sense.

⁷We abstain from using available information on the cause of death as the measure disregards that the primary cause of death may in fact be a comorbidity of the blood clot condition.

Table 1: Distribution of diagnoses among individuals developing a blood clot following vaccination

Diagnosis (ICD-10)	Share (%)	Number of individuals	Mortality rate (% deceased before 2023)
Acute myocardial infarction (I21)	30.6	7,027	22.0
Cerebral infarction or stroke (I63)	28.0	6,428	28.2
Phlebitis and thrombophlebitis (I80)	21.0	4,820	14.9
Pulmonary embolism (I26)	15.3	3,514	33.5
Other venous embolism and thrombosis (I82)	2.4	551	29.8
Arterial embolism and thrombosis (I74)	1.9	436	31.0
Portal vein thrombosis (I81)	0.8	184	49.0
Total	100	22,957	100

Notes: This table displays the prevalence of selected blood clot indications in the final sample of individuals living in Sweden in 2021-2023 (column 1 & 2). Diagnoses are included if recorded within 42 days of the first or second COVID-19 vaccination. Column 3 presents a measure of all-cause mortality for each diagnosis code referring to fatal cases within each category.

addition had serious adverse events for blood clot-like symptoms reported by medical professionals. The average age among individuals developing blood clot after COVID-19 vaccination is 74 years. The number of treated units is split equal between the first and the second dose (consistent with blood clot being unrelated to vaccination and showing no trend across time).

In Figure 1 we display the number of first-time blood clot diagnoses per month between 2019 and 2023. Ex post, there was no aggregate increase in blood clot incidence when COVID-19 vaccination was rolled out at scale. Blood clots are also common: each year, about 0.5% of individuals experience a first-time diagnosis. As a result, many blood clots occurring shortly after vaccination would mechanically be expected to be coincidental rather than vaccine-induced.

Ex ante, individuals could nevertheless worry that rare vaccine-related cases were underdetected or not yet reported. By March 2021, however, millions of AstraZeneca doses had already been administered worldwide and only a small number of suspected cases had been identified. Even allowing for reporting delays, contemporaneous evidence implied a very low probability that a newly developed blood clot was vaccine-induced—though this is precisely the kind of base-rate information that is difficult for laypeople to interpret in real time.

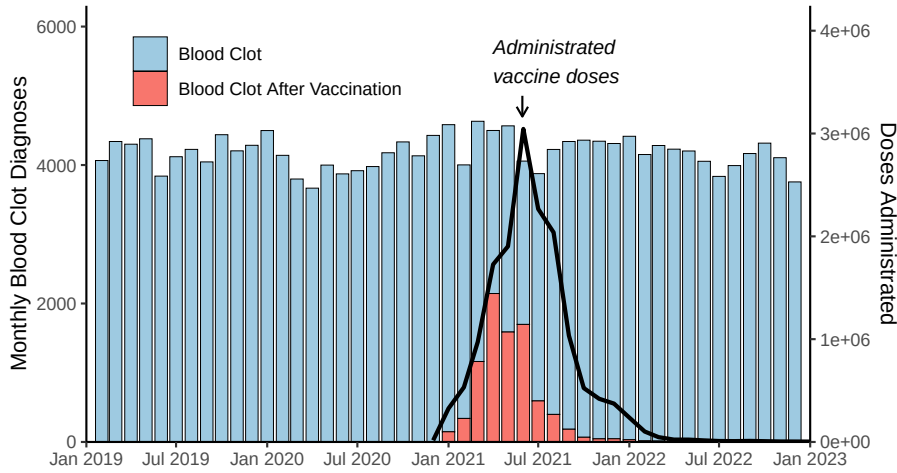


Figure 1: Timeline of Blood Clot Diagnoses

Notes: This figure displays first time blood clot diagnoses across time between 2019 and 2022 (left axis). The blue bars represent the total count of diagnoses. The red bars represents individuals that developed blood clot within six weeks after receiving either their first or second COVID-19 vaccine dose and hence constitute our main treatment group. The black line represents the monthly number of first and second administrated COVID-19 vaccine doses (right axis).

Other conditions Apart from blood clot, we construct a group of *physical conditions*, consisting of all ICD-10 S-codes (Chapter XIX: Injury, poisoning and certain other consequences of external causes), which capture acute physical injuries to specific body regions. These diagnoses are unlikely to be attributed to the COVID-19 vaccine. We expect any effects among individuals with these conditions to, if anything, reflect an “incapacitation effect”, e.g. the physical restrictions from breaking a leg make it more difficult get another dose of COVID-19. Finally, we construct a set of all *other severe acute conditions* consisting of all acute, non-physical conditions. These are similar in severity to blood clot and it is plausible that a lay person would have attributed the condition to the vaccine to a similar extent as blood clot, had blood clot not been made especially salient in the media. See Section D.1 for a more detailed description of how we select diagnoses for this category. In summary, we distinguish between three sets of diagnoses: (i) blood clots, which may be conflated with VITT; (ii) physical conditions, which are primarily incapacitating and unlikely to be attributed to the vaccine; and (iii) other severe acute conditions, which are comparable in severity to blood clots and could plausibly be attributed to the vaccine in the absence of heightened media salience.

Outcome variables We exploit the information on vaccination doses to define an outcome variable of subsequent vaccination, equal to one if an individual takes at least one more COVID-19 dose before March 2023, which marks the end of our data but also the end of the acute phase of the COVID-19 pandemic, independent of the brand of that next dose. We also study *switching*, defined as 1 if an individual continues vaccination but with a different brand than the brand that preceded the blood clot.

4 Conceptual Framework

Motivation This section presents a simple model of how vaccination choices are impacted by the experience of perceived side effects. Individuals begin by choosing whether to get an initial vaccine dose, weighing benefits against the perceived probability and severity of side effects. Following the initial dose, they may experience a health event. Drawing on the type of illness, the time from vaccination to onset, and possible links drawn between vaccination and the symptoms by media and doctors, the individual estimates a probability that their ailment was caused by the vaccine. They then update their risk assessment of the vaccine before deciding whether to take the next dose.

The model, while simple, delivers predictions about which kinds of ailments and information treatments cause individuals to drop out of the vaccination program, as well as which individuals are most responsive. We will bring these predictions to the data.

Learning about vaccine risk Individuals are uncertain about the true probability of serious vaccine side effects $\theta \in [0, 1]$. Before vaccination, their belief over θ is captured by a shared prior $p(\theta)$. After receiving the first dose, individuals may or may not experience a vaccine-induced illness— $z_i \in \{0, 1\}$ —where $z_i = 1$ with probability θ .

Importantly, z_i is unobserved by the individual. Instead, she observes a health signal y_i . The signal consists of (i) symptoms (or lack thereof), (ii) time from vaccination to onset, and (iii) cues from media or clinicians about the vaccine-illness link. The distribution of y_i depends on whether the illness was truly vaccine-induced ($z_i = 1$) or not ($z_i = 0$). We denote the respective conditional likelihood functions by g_1 and g_0 . g_0 can be thought of as the distribution of health outcomes absent vaccination, whereas g_1 represents the individual’s view of how vaccine side effects tend to present.⁸

The individual uses the health signal y_i to form a posterior distribution over θ :

$$p(\theta|y_i) \propto p(\theta) \cdot (\theta \cdot g_1(y_i) + (1 - \theta) \cdot g_0(y_i)) \quad (1)$$

which is then used to decide whether to take the next dose.

Vaccination choice Beliefs about side effect risks, weighed against benefits, are used in vaccination choices. We denote the perceived benefits of vaccination by $B(x_i)$, where x_i is a vector of observable traits. These benefits include both health and non-health considerations.

The costs of vaccination consist of the perceived risk of side effect, $E[\theta]$, multiplied by a shared severity S . For the first dose, the expectation over θ is computed with regards to the prior distribution $p(\theta)$, whereas for the second, it is computed from the posterior $p(\theta|y_i)$. Finally, we allow for an idiosyncratic shock ε_i to account for attitudes toward vaccination not captured elsewhere. The perceived net benefit of vaccination is

$$u_i = B(x_i) - E[\theta] \cdot S + \varepsilon_i \quad (2)$$

⁸Concretely, let $y_i \in \{\emptyset\} \cup \mathcal{Y}$ where $\mathcal{Y}$ is the set of health events and $\emptyset$ denotes remaining healthy. Then $g_0(\emptyset)$ will be relatively high whereas $g_1(\emptyset) \approx 0$. The degree to which an event y is attributed to the vaccine is captured by the likelihood ratio $g_1(y)/g_0(y)$. The ratio is high for symptoms occurring soon after vaccination or matching the individual’s prior about how side effect present. It is low for illnesses with delayed onset or no plausible causal link to the vaccine— $g_1(\text{broken arm}) \approx 0$.

We assume that individuals choose not to vaccinate if $u_i \leq 0$. If $u_i > 0$, individual i gets vaccinated t_i days after the vaccine is made available to them, where $t_i|u_i \sim G(\cdot|u_i)$ such that higher u_i implies earlier vaccination in distribution⁹.

Model predictions The model delivers five predictions that we bring to the data. Predictions 1–3 relate to which kinds of health signals trigger vaccination dropout, whereas 4 and 5 concern who responds more.

1. *Attribution*. Health events following vaccination reduce subsequent doses only if the symptoms can plausibly be attributed to the vaccine.
2. *Timing*. A shorter time from vaccination to symptom onset increases the effect.
3. *Salience and cues*. Effects are larger when medical professionals or public discourse reinforce the perceived link between the symptoms and the vaccine.
4. *Benefit gradient*. Individuals with lower predicted benefit from the vaccine respond more to health events.
5. *Hesitancy amplification*. Individuals who delay their first vaccination reduce their uptake of the next dose more in response to a health event.

The last two predictions share a common logic. Among those who took the first dose, individuals with lower $B(x_i)$ and/or higher t_i have lower u_i —closer to the threshold $u_i = 0$ —and are therefore more likely to be tipped into dropping out upon suffering a negative health event.

Health literacy An additional implication concerns heterogeneity by health literacy, which we operationalize as the ability to distinguish between illnesses likely to be caused by the vaccine and background health events. To capture this, let

$$g_0(y) = \begin{cases} p_0, & \text{if } y = \emptyset, \\ (1 - p_0) h_0(y), & \text{if } y \in \mathcal{Y}. \end{cases} \quad (3)$$

$$g_1(y) = \ell_i \cdot h_1(y) + (1 - \ell_i) \cdot h_0(y), \quad (4)$$

where h_0 is the distribution of non-vaccine-induced illnesses, h_1 that of vaccine-induced side effects, and $p_0 \triangleq \Pr(y_i = \emptyset | z_i = 0)$. Higher ℓ_i makes individuals more able to discern true side effects from background events, making them respond more to the former and less to the latter.

5 Method

Challenges to Identification & Matching Our empirical strategy is based on comparing immunization outcomes among individuals who developed blood clot shortly after receiving the COVID-19 vaccine to individuals who received the same vaccine but did not develop a blood clot. This approach relies on the assumption that conditional on observable characteristics, developing blood clot is orthogonal to vaccine hesitancy. We assume that this conditional independence assumption holds also among individuals who were diagnosed and additionally have their doctor report the blood clot event as a severe side effect. The main

⁹Formally, for any u, u' : $u' > u \implies G(\cdot|u)$ stochastically dominates $G(\cdot|u')$.

threats to our identification strategy are that individuals may be differentially likely to (1) develop a blood clot in the first place, (2) get diagnosed with a blood clot and (3), conditional on being diagnosed with a blood clot, have their doctor report it as a side effect. In particular, individuals may be selected in terms of predispositions affecting the probability of developing blood clot in the first place as well the inclination to consult health care professionals. Furthermore, we cannot rule out that individuals who have a latent hesitancy to vaccinate press medical professionals to report their condition as a side effect.

To handle these endogeneity concerns, we deploy a two-step matching procedure. The key idea is to identify individuals who were similar along observable characteristics by the time an individuals is diagnosed with blood clot. To that end, we first identify control individuals in terms of the COVID-19 *vaccine history* including timing and the brand of previous vaccine doses; An individual that develops a blood clot after the second dose is matched to an individual who received the same vaccine brand within the same month for the first and second dose. By matching on vaccine brand together with vaccination timing, we address what is perhaps the most important concern: individuals who are more worried about side effects selecting into vaccines perceived as safer and receive their doses later. We, furthermore, restrict the matching to individuals who by the time of developing blood clot, had not yet received their next dose. Within these vaccine history-cells, we deploy extreme gradient boosting to compute propensity scores along basic socioeconomic characteristics. Each treated unit is matched to their nearest untreated propensity score neighbor who had not yet taken their next dose.

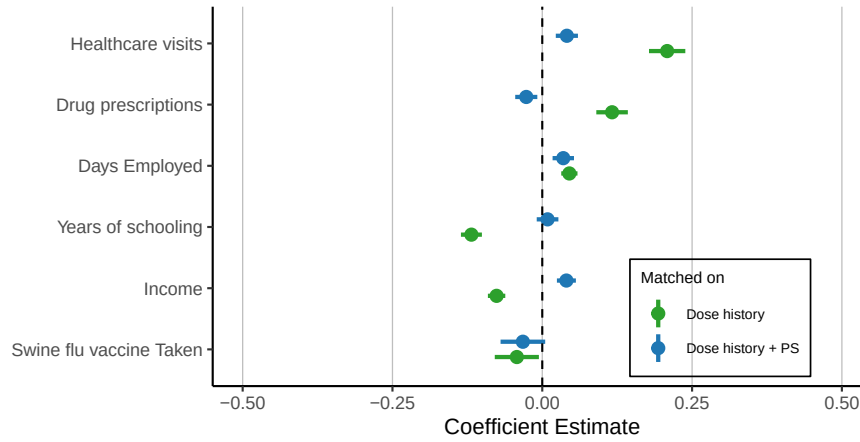


Figure 2: Balance in Pre-treatment Characteristics

Notes: This figure displays results from univariate regressions where the treatment (diagnosed with blood clot) is regressed on standardized covariates in the "Dose history" sample where we match on (i) vaccine brand(s) received, (ii) calendar month. Furthermore, each treated individual is matched to a control individual who had not yet taken their next dose by the time the treated individual is diagnosed with the blood clot. "Dose history + PS-matching" is instead based on a sample where to each treated individual we match a control individual based on propensity scores.

As evident from Figure 2, matching solely on vaccine dose history leaves residual imbalance in terms of individual health characteristics as measured through healthcare visits and drug prescriptions. Notably, while there is little difference in terms of socioeconomic characteristics, individuals in the treatment group appear to be sicker as measured by the number of drug prescriptions and healthcare visits. To approximate

conditional independence, we match individuals on observable socioeconomic characteristics that jointly predict both vaccine hesitancy and blood clot onset. The propensity score matching does a good job at achieving balance, although some residual balance remains.

6 Results

6.1 The role of media salience

In Figure 3 we display our main results for the different blood clot conditions along with estimates for physical conditions and other severe acute conditions.

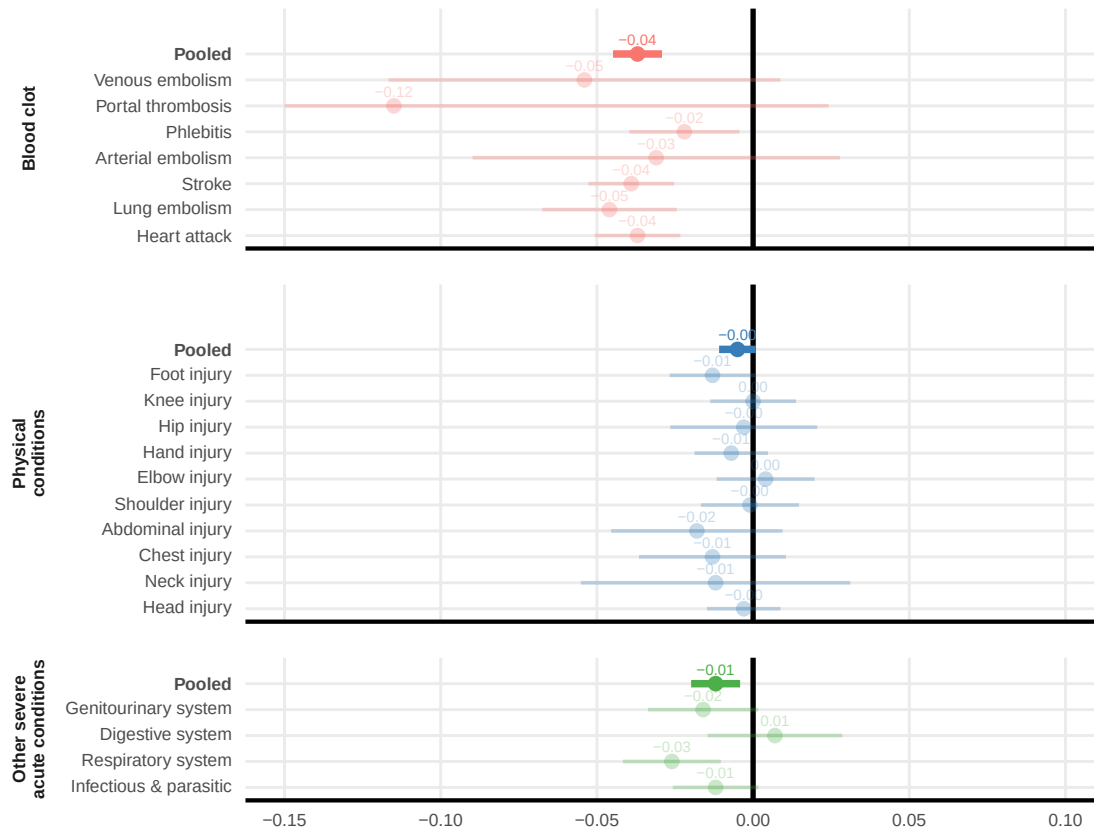


Figure 3: Vaccine Hesitancy and Media Exposure

Notes: This figure displays coefficient estimates of regressions run on matched sample where the dependent variable is equal to 1 if an individual takes additional COVID-19 vaccine doses and 0 otherwise. We consider three types of conditions: (i) Blood clot conditions that were unlikely to have been vaccine-induced but plausibly confounded to have been induced by the COVID-19 vaccine; (ii) Physical conditions that were highly unlikely to have been adverse events and where the individual is unlikely to infer that they were adverse events; (iii) Other Severe Acute conditions that were ex-ante equally likely to have been adverse events from the COVID-19 vaccine but turned out to be unlikely to have been vaccine-induced and where there was little to no public discussion revolving them being potential adverse event.

We find an average reduction in uptake of 4 percentage points among individuals who experienced a blood

clot soon after vaccination.¹⁰ There is some heterogeneity across specific blood clot diagnoses, although our data does not allow for sharp conclusions on this point. Most remarkably, the diagnoses presenting most similar to VITT (venous embolism and portal thrombosis), as per the location of the clotting, return the largest estimates.

We find negative effects also for other severe acute conditions, for instance relating to the digestive system or the respiratory system, albeit smaller in magnitude and mostly statistically insignificant. This difference is unlikely to reflect differences in severity: as shown in Table D1, these conditions are also severe and associated with high mortality rates. Taken together, we find evidence of a small degree of misattribution stemming from developing a disease soon after vaccination in itself but the media coverage of rare and specific blood clots amplified hesitancy among individuals who experienced similar events shortly after vaccination.

We do not detect an effect for individuals experiencing a pure placebo diagnosis, such as a foot injury. This is in line with the notion that individuals can not plausibly connect their diagnosis to the vaccine and therefore do not adjust their beliefs towards the vaccine.

Spillover to family members To understand the impact on vaccine hesitancy of an additional blood clot case we turn to studying spillovers on personal networks. In Table B1, we report the estimates for partners, siblings, and children of the diseased. The outcome variable is defined analogously to our previous approach, equaling 1 if a network member takes another vaccine dose after the focal individual within their family is diagnosed with a blood clot, and 0 otherwise. We report our estimates for the case that the focal individual died or lived on after experiencing a blood clot. Note that death is coded irrespective of cause. Overall, the estimated effects are small and negative, although the estimates for children are close to 0. Notably, the effect for partners of individuals with a non-fatal diagnosis is roughly half the size of the effect on the diseased themselves. The estimates for siblings are smaller in size potentially capturing that partners are more exposed to the struggles accompanying the diagnosis than other network members. This pattern suggests that the responses we document are not solely driven by individuals updating beliefs about their own predisposition to side effects, but more broadly by the inference that side effects are relatively common when a family member develops a blood clot shortly after vaccination. Note, that we find insignificant and small estimates for all but siblings in cases where the focal individual dies. Overall, our results bear relevance in the more general quantification of social costs of side effects. Close family learns from the mere observation - and exposure to personal narrative - of a blood clot condition surfacing post vaccination. These results relate to Miltner and Riberth (2026), where we find that close family members of individuals developing narcolepsy post their swine-flu vaccination were significantly less likely to be vaccinated during the COVID-19 vaccine.

Accounting for the spillovers to family members, the coefficient point estimates in Table B1 suggest that every blood clot is associated with 0.08 individuals dropping out of the vaccination schedule¹¹. In other

¹⁰In an alternative specification, we explicitly construct a control group where control units are drawn from a donor pool of individuals who developed blood clot *before* the COVID-19 pandemic. This approach directly creates comparable units in terms of propensity to develop blood clot. However, this group of individuals is smaller rendering it less feasible to target other imbalances in other characteristics such as socioeconomic characteristics. Either way, the approach yields identical results for blood clot diagnoses, see Figure B4.

¹¹This number comes from multiplying the estimated effect sizes in Table B1 with the size of each network among the treated units

words, about half of the effect on stopping vaccination comes from the affected individual, and the other half comes from family members stopping vaccination.

6.2 What information is used?

Heterogeneity by time until blood clot Our explanation for why individuals become more hesitant after experiencing blood clot is that they form an assessment about the risk of developing side effects from the next dose. This in turn depends on an assessment of the likelihood that the blood clot that they developed was induced by their previous vaccine dose. The perceived link should be stronger for individuals who developed blood clot or any plausibly vaccine-induced condition closer in time to vaccination. We illustrate the results to testing this hypothesis in Figure 4, where we split the sample by the number of weeks elapsed between the COVID-19 vaccination and the blood clot incident. In Figure B5 we display the corresponding results for the decision to switch vaccine brand rather than abstaining from vaccination completely.

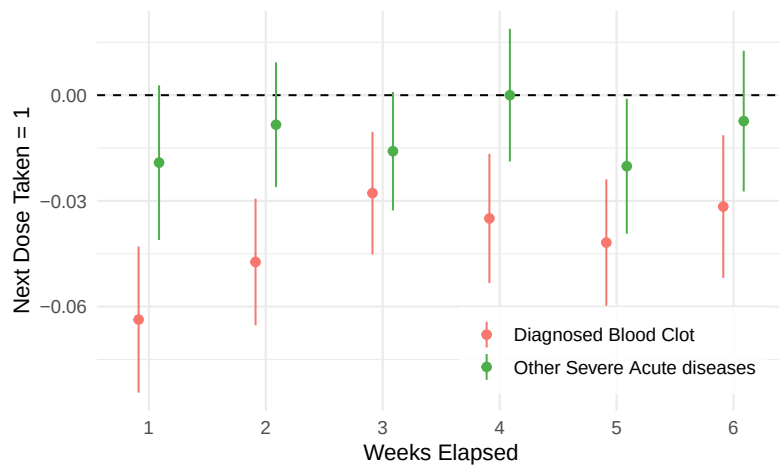


Figure 4: Heterogeneity by Time Elapsed Between Vaccine and blood Clot

Notes: This figure displays coefficient estimates of regressions run on matched sample where the dependent variable is equal to 1 if an individual takes additional COVID-19 vaccine doses and 0 otherwise. The sample is split by the number of weeks elapsed between the last COVID-19 vaccine dose and developing a condition.

We find that the estimates for blood clots are decreasing with latency period, suggesting individuals developing a blood clot condition long after vaccination are less hesitant to proceed with their vaccination schedule. If anything, there is less of a negative gradient for ex-ante plausible conditions, suggesting that for conditions that appeared even less plausibly linked to vaccination at the time (although all were very unlikely ex post), individuals who inferred a vaccine connection did so regardless of timing. Taken together, individuals appear somewhat rational in the sense that they are less likely to attribute a blood clot to the vaccine when it occurs long after vaccination. At the same time, the overall pattern supports the interpretation that the effects are driven by misattribution.

(2.33 children, 2.1 siblings and 1 partner)

The role of doctors' communication A key question is whether misattribution arises merely from experiencing a negative health event, or whether it requires additional confirmation from external sources. To shed further light on this, we consider effects among a subset of individuals who, on top of developing blood clot after COVID-19 vaccination, had it reported as a suspected side effect by a healthcare professional. Under the premise that doctors communicate their decision to report to the patient, we view doctors' reporting as confirming to the patient that the condition was a side effect. We find significantly larger effects in this subset with an average effect of 16 percentage points compared to the effect of just developing blood clot (4 percentage points). Our interpretation is that doctors report it as a side effect and in doing so cues the patient perception that the condition was vaccine-induced, increasing the perceived probability that the individual will develop side effects also from subsequent doses.¹²

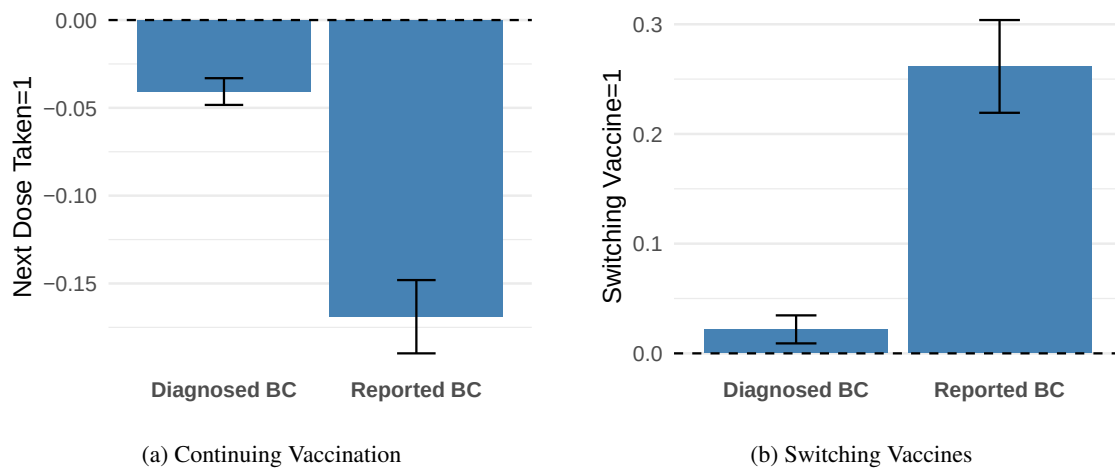


Figure 5: Role of doctors' communication

Notes: This figure reports coefficient estimates from regressions estimated on the matched sample. Diagnosed BC indicates individuals diagnosed with a blood clot within six weeks of receiving a COVID-19 vaccine (or their matched controls), while Reported BC further restricts the sample to cases reported as suspected adverse events by a healthcare professional. **Panel (a)** uses an indicator for continued COVID-19 vaccination as the dependent variable. **Panel (b)** uses an indicator for switching vaccine brand as the dependent variable.

These results speak to an important policy trade-off. While doctors need to report suspected side effects to detect new ones, the act of reporting side effects also cause individuals to become more hesitant, despite the fact that the experienced symptom was unlikely to have been a side effect.

Heterogeneity across vaccine brands We exploit heterogeneity across vaccine brands to study whether individuals' responses to the VITT episode were narrowly targeted to AstraZeneca—the vaccine associated with VITT—or whether the shock spilled over to other brands as well. Spillovers would be consistent with imperfect attribution and “generalization” in risk beliefs, implying that the social costs of salient adverse events may extend beyond closely related conditions and also affect vaccines that were not implicated. To this end, we revisit our analysis across the three main vaccine brands administered in Sweden. We present

¹²One might expect stronger responses among women because early reporting around the AstraZeneca–blood clot episode often emphasized cases in younger women, potentially making the side effect narrative more salient for females. Figure B6 shows no evidence of such gender heterogeneity in either continuation or switching behavior.

the results of this exercise in Figure 6. To account for compositional differences in who receive the different brands, we reweight observations by age, risk group status and whether or not they have a doctor in their family following DiNardo et al. (1996). We display these results in Figure B3.

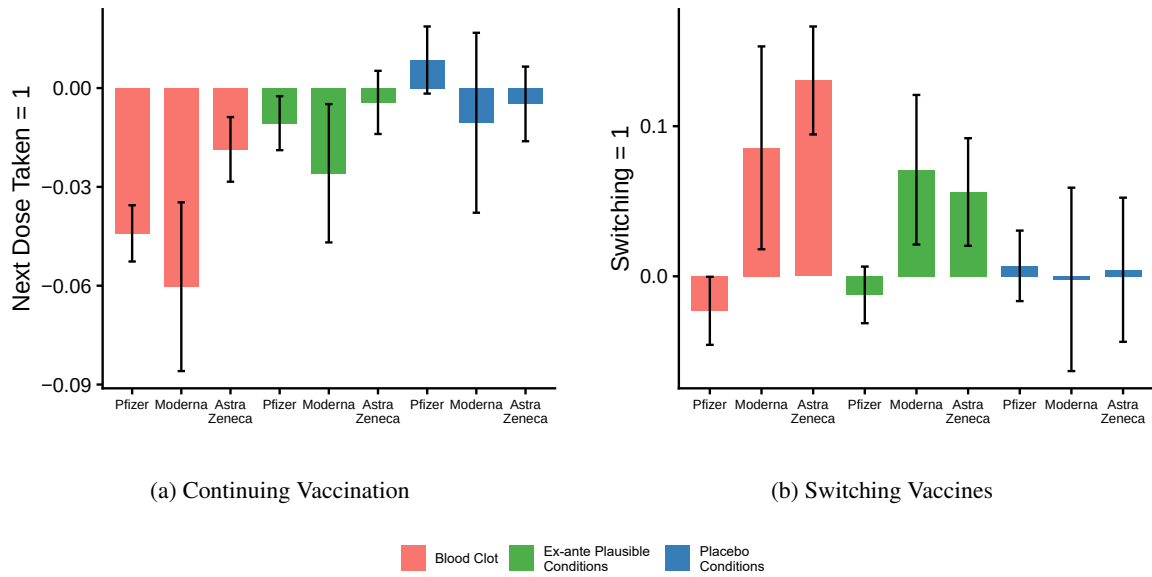


Figure 6: Brand Heterogeneity

Notes: This figure displays coefficient estimates of regressions run on matched sample where the dependent variable is equal to 1 if an individual takes additional COVID-19 vaccine doses and 0 otherwise. Coefficient estimates are split up by the brand that the individual received prior to developing blood clot.

For the decision to take the next dose, there is little heterogeneity across brands, suggesting that individuals who drop out of the vaccine schedule do not strongly distinguish which vaccine might have caused the adverse event.

However, many individuals respond by continuing vaccination but switching brand. Experiencing a blood clot after vaccination with AstraZeneca or Moderna significantly increases the probability of switching, with the largest effects for AstraZeneca. We do not detect comparable switching after a blood clot following Pfizer, consistent with Pfizer being perceived as the safest: while abstention effects look similar, the incentive to switch away from Pfizer is smaller.

Finally, switching effects are larger for AstraZeneca in the blood-clot analysis than in the ex-ante plausible analysis, whereas the difference is smaller for Moderna. This pattern is suggestive that media exposure played a role in amplifying hesitancy and switching around AstraZeneca.

6.3 Who is affected?

Heterogeneity in benefits of vaccination Abstaining from vaccination entails an excess risk of falling ill from COVID-19. Do individuals at higher risk of severe COVID-19 respond less to exposure to side effects? In Figure 7, we use two proxies for perceived benefits of vaccination: age and the time elapsed between when the vaccine became available to an individual and the timing of the first dose. We find that older individuals, and those who took the dose preceding the blood clot earlier, react less than younger individuals or those

who took their first dose later.

For age, older individuals face a higher risk of severe COVID-19 and therefore have greater benefits from vaccination than younger individuals, who are on average more indifferent between vaccinating and not vaccinating in the absence of blood clot. A similar pattern holds for time since availability. Individuals who take the vaccine shortly after it becomes available are likely those with higher perceived benefits and are therefore less likely to stop immunization, compared to individuals who delay vaccination, for whom a shock to perceived costs is more likely to tilt the decision toward abstaining from future vaccination.

Combining the results from Figure 7 panel (a) and panel (b), there is much less of a gradient for other Severe Acute Diseases compared to blood clot, suggesting that these individuals do not trade off benefits and costs in the same way as those experiencing blood clot. As previously stated, the most plausible explanation is that the small negative effects that we find largely reflects an incapacitation effect that is independent of age or time until first dose.

Taken together, these results suggest that individuals trade off the benefits and costs of vaccination. While the costs, in the form of perceived side-effect risk, are similar across individuals experiencing blood clot, perceived benefits vary, generating heterogeneity in the decision to stop vaccinating.

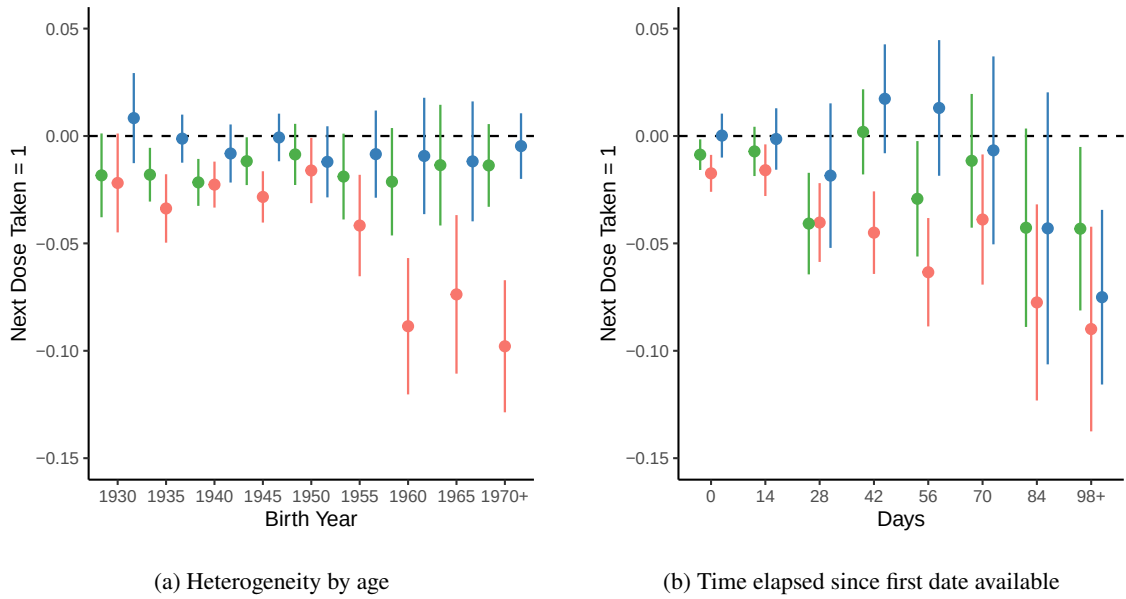


Figure 7: Eliciting the perceived benefits of vaccination

Notes: This figure displays coefficient estimates of regressions run on matched sample where the dependent variable is equal to 1 if an individual takes additional COVID-19 vaccine doses and 0 otherwise. **Panel (a):** Coefficient estimates split up by birth year. **Panel (b)** Coefficient estimates split up by weeks elapsed since first date the COVID-19 vaccine is available. Date of first availability is defined in Section D.2.

Heterogeneity in health literacy Finally, we are interested in heterogeneity by personal health literacy. This allows us to pin down the role of understanding cues, i.e. media communication on a rare side effect such as VITT, in the observed effects on vaccine adherence. For instance, having a doctor in the family may make it easier to understand the lack of clinical association between general blood clot conditions and the rare cases of VITT. Figure 8 reveals our main estimates by different measure of health literacy. We find

no statistically significant difference with respect to health literacy, but if anything the coefficients suggest that the blood clot-signal cause individuals with high health literacy to update less and hence react less to experiencing blood clot.

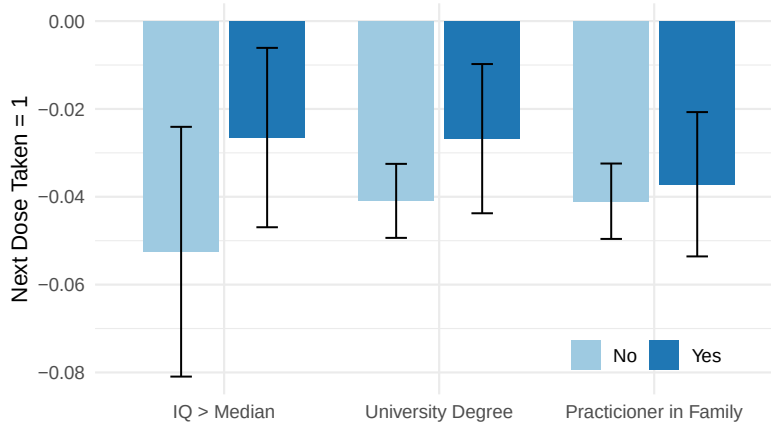


Figure 8: Heterogeneity in Health Literacy

Notes: This figure displays coefficient estimates of regressions run on matched sample where the dependent variable is equal to 1 if an individual takes additional COVID-19 vaccine doses and 0 otherwise. We split up three proxies for health literacy. *Doctor in family* is defined as “Yes” if an individual has a parent or a sibling with a medical degree or a nursing degree. *High Cognitive Ability* is based on cognitive tests completed by military conscripts. It is equal to “Yes” if an individual had above median score on the test compared to peers born the same year. *University Degree* is equal to “Yes” if an individual has at least a bachelor degree, corresponding to three years of higher education.

7 Conclusions

Is the communication of side effects costly? This paper shows that experiencing an adverse health event after vaccination can lead individuals to falsely attribute symptoms to the vaccine. Informational cues, such as salient media attention around side effects or a clinician’s decision to file a side-effect report, drive this response. In our setting, experiencing a blood clot—clinically unrelated to the rare vaccine-induced condition—shortly after COVID-19 vaccination reduces the probability of continuing the vaccine schedule. This pattern is consistent with individuals generalizing from the most salient side-effect narrative to their own symptoms. By contrast, we detect no meaningful change in vaccination behavior after conditions that present differently from the rare vaccine-induced blood clots, highlighting the role of media coverage on the interpretation of symptoms following vaccination.

Theoretically, our results highlight how people generalize information across health domains: effects spill over not only to blood-clot diagnoses that differ from the specific adverse event of concern, but also across vaccine brands, even when only one brand is linked to elevated risk. Taken together, this points to substantial misattribution, with individuals extrapolating from a rare, brand-specific adverse event to other blood-clot diagnoses and to vaccines for which no elevated risk was suspected.

From a policy perspective, the results suggest that the communication about side effects bears potential

risk for vaccine campaigns. The costs arise as soon as news about a potential side effect breaks: the withdrawal of the AstraZeneca vaccine shortly thereafter did not prevent a persistent increase in hesitancy among those who experienced blood clots in the same period. The evidence on clinician side-effect reporting also points to a similar trade-off. Reporting is essential for detecting genuine adverse reactions, yet the act of reporting may unintentionally reinforce patients' beliefs that their condition was vaccine-induced.

The practical scope of our findings can be substantial. In the MMR–autism case, a highly salient medical claim was later retracted, yet many people did not fully revise their beliefs. Instead, vaccine confidence remained lower for years in that context ([Miltner and Riberth, 2026](#)). Evidence of such durable effects, even after the underlying claim was discredited, suggests that the AstraZeneca–blood clot episode may likewise generate persistent trust impacts when early safety signals are later shown to be false or overstated.

Public debates about vaccine side effects are not uncommon, but which risks become “the” public story can be somewhat arbitrary. For example, myocarditis and pericarditis were identified as very rare adverse events following the mRNA vaccines, concentrated among young males, yet this risk did not generate the same sustained debate as the AstraZeneca–blood clot episode. This suggests that media, clinicians, and other stakeholders, shape which risks are amplified in the public perception, and that salient outcomes such as blood clots may be especially likely to dominate attention. The lesson for risk communication is that when attention is driven by salience as much as by incidence, some risks can be amplified out of proportion while other clinically important risks remain underweighted. Future research should therefore focus on how these narratives form and spread in increasingly digital information environments, and on how institutions can communicate safety signals in ways that preserve trust while keeping perceived risks aligned with the underlying evidence.

References

- Agosti, F., V. Toffolutti, N. Cavalli, S. Nivakoski, M. Mascherini, and A. Aassve (2022). Information and vaccine hesitancy: Evidence from the early stage of the vaccine roll-out in 28 european countries. *Plos one* 17(9), e0273555.
- Asan, L., J. Kleine-Borgmann, B. Bozkurt, B. Frank, M. Köhrmann, C. Kleinschnitz, and U. Bingel (2024). Media coverage of covid-19 vaccination-associated cerebral venous sinus thrombosis was followed by a surge in emergency presentations due to headache—observations from a university hospital in germany. *Frontiers in Psychiatry* 15, 1378472.
- Ashraf, N., G. Bryan, A. Delfino, E. Holmes, L. Iacovone, C. Meyer, and A. Pope (2024). Learning to see a world of opportunities. Technical report, Working Paper. Accessed November 22, 2024. <https://navaashraf.com/wp...>
- Bordalo, P., J. J. Conlon, N. Gennaioli, S. Y. Kwon, and A. Shleifer (2022, 08). Memory and probability*. *The Quarterly Journal of Economics* 138(1), 265–311.
- Deiana, C., A. Geraci, G. Mazzearella, and F. Sabatini (2022). Perceived risk and vaccine hesitancy: Quasi-experimental evidence from italy. *Health Economics* 31(6), 1266–1275.
- DiNardo, J., N. M. Fortin, and T. Lemieux (1996). Labor market institutions and the distribution of wages, 1973-1992: A semiparametric approach. *Econometrica* 64(5), 1001–1044.
- Espín-Sánchez, J.-A., S. Gil-Guirado, and N. Ryan (2023, June). Praying for rain. NBER Working Paper 31411, National Bureau of Economic Research.
- García-Abeijón, P., C. Costa, M. Taracido, M. T. Herdeiro, C. Torre, and A. Figueiras (2023, July). Factors associated with underreporting of adverse drug reactions by health care professionals: A systematic review update. *Drug Safety* 46(7), 625–636. Epub 2023 Jun 6.
- Haas, J. W., F. L. Bender, S. Ballou, J. M. Kelley, M. Wilhelm, F. G. Miller, W. Rief, and T. J. Kaptchuk (2022). Frequency of adverse events in the placebo arms of covid-19 vaccine trials: a systematic review and meta-analysis. *JAMA network open* 5(1), e2143955–e2143955.
- Hazell, L. and S. A. Shakir (2006). Under-reporting of adverse drug reactions: a systematic review. *Drug Safety* 29(5), 385–396.
- Johansson, M., L. Johansson, and M. Lind (2014). Incidence of venous thromboembolism in northern sweden (veins): a population-based study. *Thrombosis journal* 12(1), 6.
- Jones, S. P. (2020, April). Imperial college london yougov covid data hub. <https://www.coviddatahub.com>. Online resource; accessed via <https://www.coviddatahub.com>.
- Katsoularis, I., O. Fonseca-Rodríguez, P. Farrington, H. Jerndal, E. H. Lundevaller, M. Sund, K. Lindmark, and A.-M. Fors Connolly (2022). Risks of deep vein thrombosis, pulmonary embolism, and bleeding after covid-19: nationwide self-controlled cases series and matched cohort study. *BMJ* 377.

- Klok, F. A., M. Pai, M. V. Huisman, and M. Makris (2022). Vaccine-induced immune thrombotic thrombocytopenia. *The Lancet Haematology* 9(1), e73–e80.
- Malmendier, U., S. Nagel, and Z. Yan (2021). The making of hawks and doves. *Journal of Monetary Economics* 117, 19–42.
- Malmendier, U. and G. Tate (2005, November). Ceo overconfidence and corporate investment. *The Journal of Finance* 60(6), 2661–2700.
- Miltner, S. and J. Riberth (2026). Adverse health events and vaccine hesitancy. *forthcoming*. <https://jonatanriberth.github.io/jmp/jmp.pdf>.
- Zhao, L., H. Tran, M. Risk, and G. Nair (2024). Risk of blood clots after covid-19 vaccination and infection: a risk-benefit analysis. *Research square*, rs–3.

Appendix A. Medical background of vaccine induced blood clots

Vaccine-induced immune thrombotic thrombocytopenia (VITT), a form of thrombosis with thrombocytopenia syndrome (TTS), is an extremely rare adverse event that has been observed following viral vector vaccines (30-42 days), in particular AstraZeneca and Johnson & Johnson COVID-19 vaccines. No cases of VITT were documented for the mRNA vaccines Pfizer and Moderna.

The clinical presentation of VITT, like for regular blood clots events, depends on the precise location of the clot. The blood clots for VITT commonly form in unusual sites of the body (mainly cerebral or sinus vein thromboses), although different presentations are possible, i.e. a deep vein thrombosis. The critical differentiation between the two types is the development of the clots. VITT is an immune reaction to PF4 (platelet factor 4), a protein stored in blood cells, the so called platelets. In case of VITT, the body produces an immune response causing hyperactivation and clumping of PF4. For this reason, a central diagnosis criteria for VITT is *low* blood platelet count, which is in stark contrast to other blood clot conditions. Note, that the pathology is similar to Heparin-induced thrombocytopenia (HIT), a condition of low platelets that is also associated with blood clots, caused by the paradox physical reaction to the blood thinner heparin. Medical practitioners are able to identify the count of blood platelets in laboratory testing. In the absence of a competing potential cause for the observation of low blood platelets, i.e. heparin administration, the diagnosis is further fostered by confirming the last date of COVID-19 vaccine.

The medical treatment of early VITT presented similar features, compared to the treatment of common blood clots. The main target of acute therapy is the prevention of further growth in existing clots and further clotting events. This is commonly achieved by administrating anticoagulant medication, blood thinners, such as heparin injections or equivalent products for oral administration. Due to the similarity in clinical presentation HIT and VITT, heparin is some times recommended to be exchanged for non-heparin based anticoagulants. Patients are monitored during acute treatment and regularly checked for several months afterwards.

Appendix B. Additional Descriptive Statistics & Additional Results

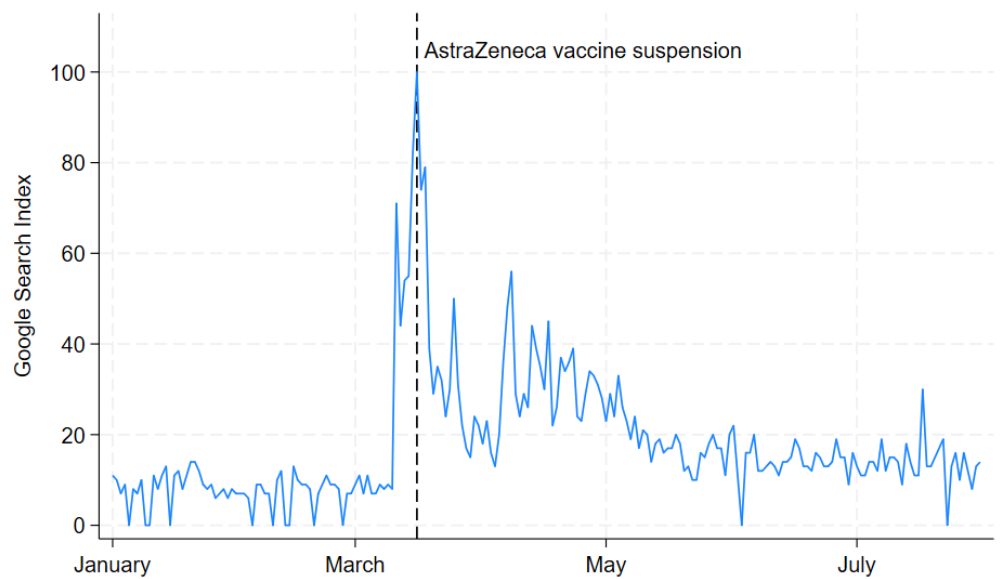


Figure B1: Timeline of Google Searches

Notes: This figure displays the daily measure of google search index for Sweden using the keyword "blodpropp" between January 1st and August 31st 2021. The frequency of searches are indexed to the time of highest search volume (= 100). The vertical dashed line represents the timing of the AstraZeneca vaccine suspension in Sweden on March 16 2021. Data last accessed January 20, 2026.

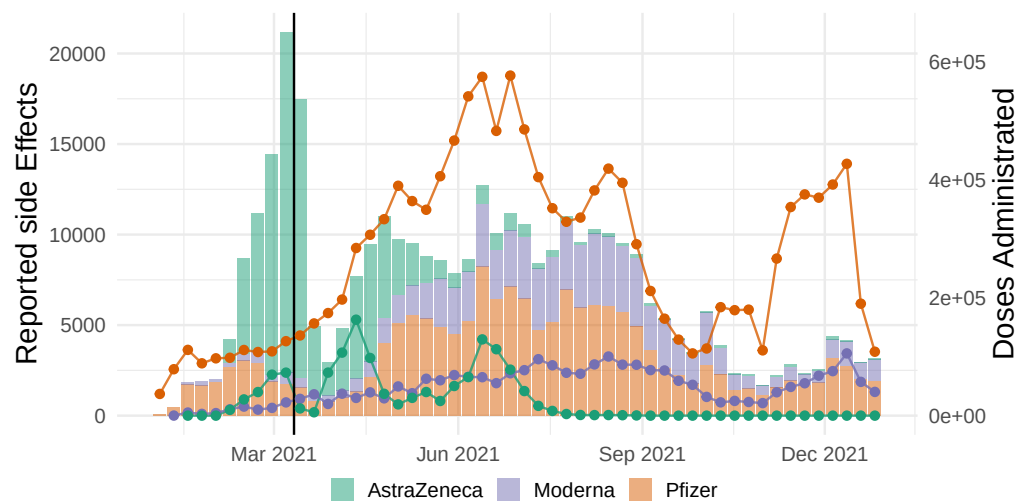


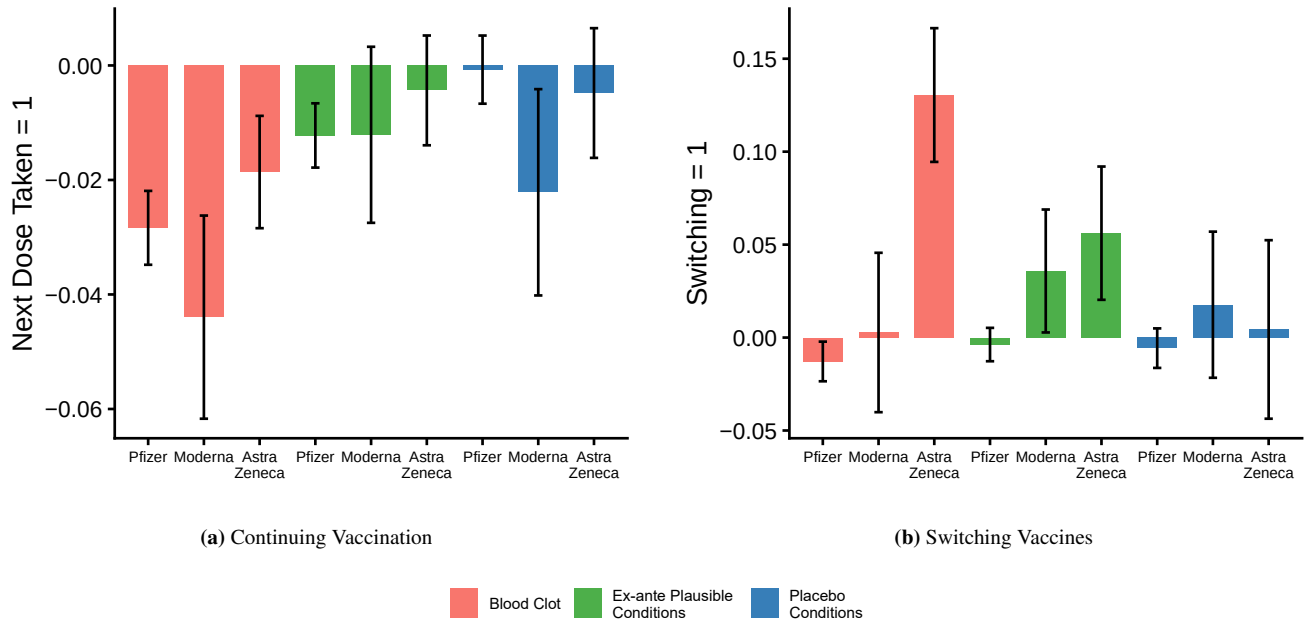
Figure B2: The Blood Clot Episode: Blood Clot Reports Over Time

Notes: This figure displays reported reported side effects for blood clot symptoms across time (bars). In addition, number of COVID-19 doses administrated are displayed (lines). The black line represents the first report of a potential link between AstraZeneca and an excess risk of developing blood clot.

Table B1: Effect of Blood Clot on Network Members' Vaccination Decisions

	Children	Partners	Siblings
<i>Focal individual died = 0</i>			
Treated	0.003	-0.019	-0.013
	(0.003)	(0.003)	(0.004)
n obs	84106	24590	59477
<i>Focal individual died = 1</i>			
Treated	0.005	-0.009	-0.018
	(0.004)	(0.006)	(0.008)
n obs	29190	6431	16155

Notes: This table reports estimates of the effect of exposure to a family member developing blood clot shortly after vaccination on the vaccination uptake of network members. Each column corresponds to a separate regression. The dependent variable is equal to 1 if an individual takes at least one COVID-19 vaccine dose after the family member is diagnosed with blood clot. Standard errors are clustered at the match-pair level.

**Figure B3:** Brand Heterogeneity, DFL reweighted

Notes: This figure displays coefficient estimates of regressions run on matched sample where the dependent variable is equal to 1 if an individual takes additional COVID-19 vaccine doses and 0 otherwise. coefficient estimates are split up by the brand that the individual received prior to developing blood clot.

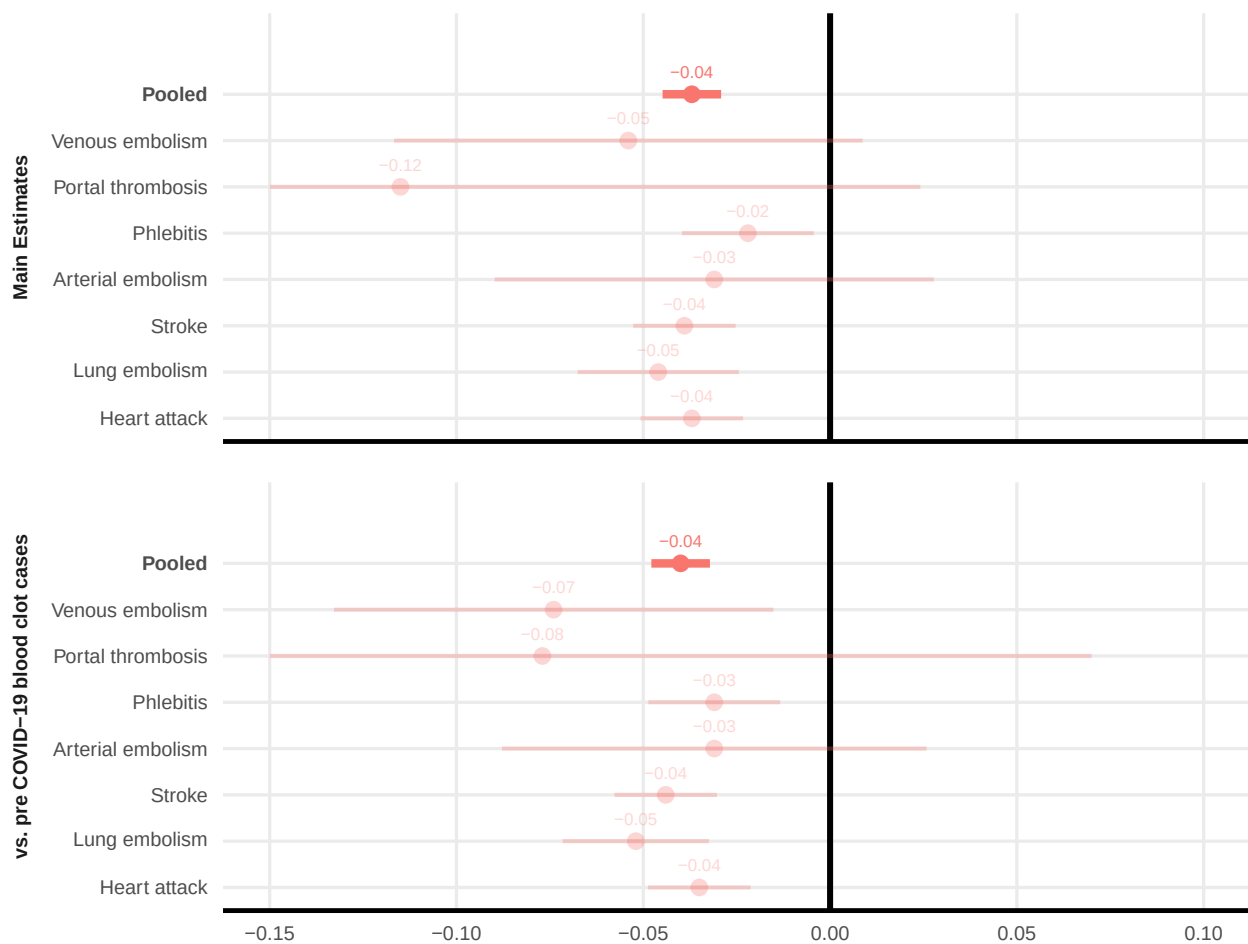


Figure B4: Vaccine Hesitancy and Media Exposure

Notes:

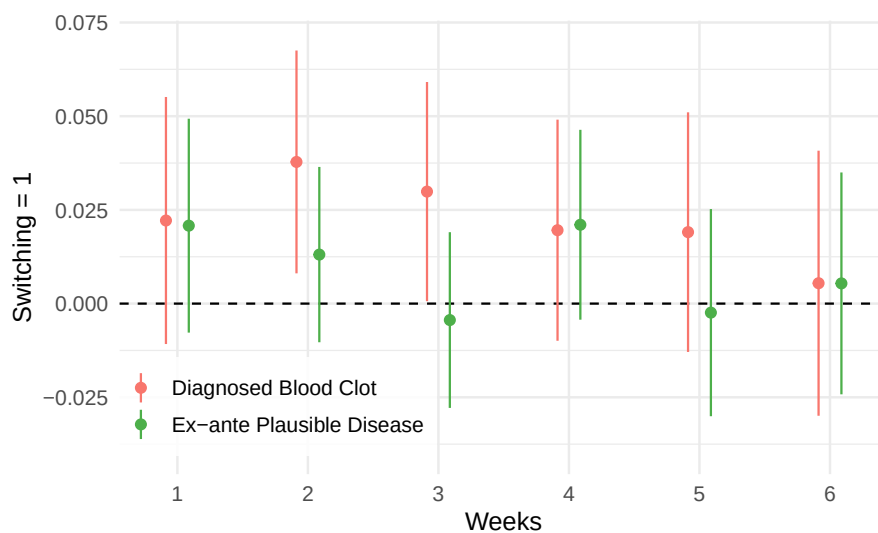


Figure B5: Heterogeneity by Time Elapsed Between Vaccine and blood Clot

Notes: This figure displays coefficient estimates of regressions run on matched sample where the dependent variable is equal to 1 if an individual takes additional COVID-19 vaccine doses and 0 otherwise. The sample is split by the number of weeks elapsed between the last COVID-19 vaccine dose and developing a condition.

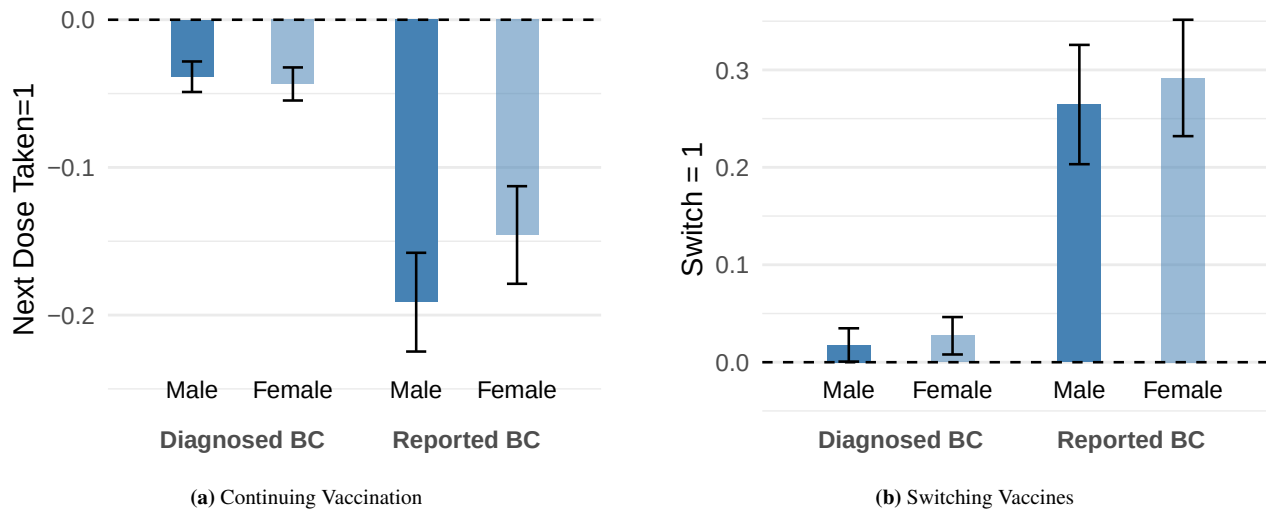


Figure B6: Role of doctors' communication

Notes: This figure reports coefficient estimates from regressions estimated on the matched sample where estimates are split up by gender. Diagnosed BC indicates individuals diagnosed with a blood clot within six weeks of receiving a COVID-19 vaccine (or their matched controls), while Reported BC further restricts the sample to cases reported as suspected adverse events by a healthcare professional. **Panel (a)** uses an indicator for continued COVID-19 vaccination as the dependent variable. **Panel (b)** uses an indicator for switching vaccine brand as the dependent variable.

Appendix C. Description of Matching

Propensity scores are computed using an XGBoost gradient boosting model with a binary logistic objective. Categorical variables are one-hot encoded, including missing-value indicators. The model is trained using 5-fold stratified cross-validation. To prevent overfitting, we employ early stopping within each fold of cross-validation after 10 rounds of no performance improvement on the test data set. Hyperparameters are set as follows: a learning rate (η) of 0.1, a maximum tree depth of 4, a minimum child weight of 1, a subsample fraction of 0.8, and a column subsampling fraction of 0.8. These settings provide moderate regularization and control overfitting while capturing nonlinear relationships.

The large number of potential matches in combination with the tight blocking conditions makes the task to find nearest neighbors without replacement challenging. Instead, we implement matching with replacement. Given the large number of candidate control individuals, this is unlikely to be a concern.

Appendix D. Data & Variable Definitions

D.1 Defining ex-ante plausible conditions

To identify ex-ante plausible conditions, we begin by selecting diagnoses that are sufficiently common to generate meaningful statistical power—those with at least 1,000 cases per year on average between 2015 and 2022 (596 ICD codes). We then restrict to diagnoses for which more than half of cases originate in specialized care (93 codes), ensuring that the conditions reflect medically significant events rather than routine primary-care contacts. Next, we exclude ICD chapters that are clearly unsuitable for our context: circulatory diseases targeted by specific treatments (chapter I), external causes of injury (chapters S–Y), non-diagnoses such as symptoms or health-status factors (chapters R, U, Z), pregnancy-related and congenital conditions (chapters O, P, Q), cancers (chapters C, D), and mental-health conditions (chapter F), leaving 31 codes. Finally, we remove a small set of individual diagnoses that are either dominated by COVID-specific patterns (e.g., certain respiratory infections), reflect chronic or lifestyle conditions (e.g., diabetes complications, COPD, nutritional deficiencies, alcohol-related liver disease), represent post-surgical complications, or are too closely related to blood clots to serve as a meaningful comparison. This results in a focused set of 19 ICD codes that plausibly could be mistaken for vaccine-related by a non-expert but are not mechanistically linked to COVID-19 vaccination.

Table D1: Distribution of diagnoses among individuals developing a blood clot following vaccination

Diagnosis (ICD-10)	Share (%)	Number of individuals	Mortality rate (% deceased before 2023)
Infectious agents (B95–B98)	27.6	9,359	24.2
Other bacterial diseases (A30–A49)	22.6	7,650	32.6
Appendicitis (K35–K38)	16.4	5,552	1.2
Acute kidney failure (N17–N19)	8.5	2,881	41.1
Bacterial pneumonia (J15)	7.1	2,415	36.0
Pleural effusion (J90)	6.0	2,040	47.0
Unspecified kidney failure (N19)	2.6	882	40.6
Peritonitis (K65–K67)	2.4	825	18.5
Respiratory failure (J96)	1.6	546	58.1
Pulmonary oedema (J81)	1.5	497	46.6
Aspiration pneumonitis (J69)	0.9	317	60.8
Pneumococcal pneumonia (J13)	0.7	248	18.4
Respiratory failure, unspecified (J96.9)	0.7	235	22.6
Hepatic failure (K72.9)	0.4	138	57.0
Pyothorax / empyema (J86)	0.3	101	22.6
Others	0.6	201	45.3
Total	100.0	33,887	26.3

D.2 Defining date of availability

We adopt a data-driven approach to determine when the vaccine first becomes available to an individual, defining availability dates separately for each birth-year-by-healthcare-region combination. For each birth-year $\times$ region cell, let $f(i)$ denote the vaccination date of the i -th individual to be vaccinated (excluding healthcare workers). We define the date of availability as $f(i^*)$, where

$$i^* = \arg \min_{i \leq N-50} \{f(i+50) - f(i)\}.$$

Intuitively, the availability date is the vaccination date at which the time span between individual i and the individual vaccinated fifty places later is minimized.